

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

Criteria	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I B.Lakshmi Lohitha(192324182) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:B.Lakshmi Lohitha

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

Q3. Dataset: Mobile App User Behavior

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

1- Hospital patient segmentation :

Patient ID	Age	monthly medical expense (₹)	Health Risk score	SSE values	
				K	SSE
P ₁	28	12,000	65	1	600
P ₂	55	25,000	40	2	350
P ₃	42	20,000	75	3	220
P ₄	30	11,000	70	4	200
P ₅	60	28,000	35	5	190

A health care optimization uses k-means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal numbers of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision making.

⇒ sse means sum of squared errors
 → we are using elbow method for predicting the values.

K	SSE	Reduction from previous K
1	600	—
2	350	250
3	220	130
4	200	20
5	190	10

when $k = 2 \text{ \& } 3$ the sse value is reduced much & when k value $= 4 \text{ \& } 5$ the decrease is very small. This variation happens after the value of k is 3. $\therefore$ Elbow method suggest k value as 3.

based on health Risk score, 3 clusters are formed

1. low risk, low expense
2. Moderate risk, High Expense
3. High Risk, High expense.

Impact on Decision making

For Low risk Group preventive care and Routine monitoring
moderate Regular checkups.
High risk Intensive monitoring with proper treatment.

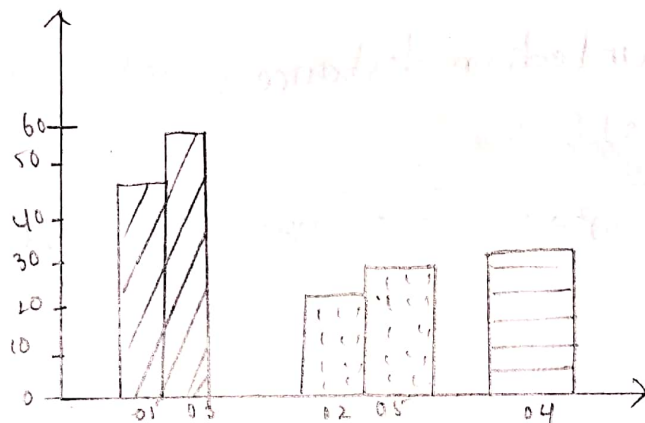
2. E-commerce order Analysis

<u>order ID</u>	<u>product category</u>	<u>order value (₹)</u>
01	electronics	65,000
02	clothing	25,000
03	electronics	57,000
04	furniture	30,000
05	clothing	27,000

A Retailer Applies Hierarchical Agglomerative clustering (HAC) to customer orders using approximately encoded Product - category info and standard order values. A dendrogram is generated to interpret how clusters are possibly formed and determine the appropriate number of clusters. by selecting a suitable case off level on the dendrogram. explain how the selected cluster can help understand customer purchasing.

HAC - calculates dis b/w clusters
 merge two closest clusters.
 Recalculates the distance
 continue until it become 1 cluster

- > 01 & 03 forms one cluster - electronics
- > 02 & 05 forms another - clothing
- > 04 furniture.



But the optimal number of clusters cannot be determined by the Given data.

1. one hot encoding:-

orders	electronics	clothing	furniture
01	1	0	0
02	0	1	0
03	1	0	0
04	0	0	1
05	0	1	0

2. Normalization (min-max)

$$x_i' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

$$x_{\min} = 25000$$

$$x_{\max} = 57000$$

$$0_1 = \frac{55000 - 25000}{57000 - 25000} = \frac{30000}{32000} = 0.9375$$

$$O_2 = \frac{25K - 25K}{57K - 25K} = 0$$

$$O_3 = \frac{57K - 25K}{57K - 25K} = 1$$

$$O_4 = \frac{30K - 25K}{57K - 25K} = \frac{5K}{32K} = 0.1562$$

$$O_5 = \frac{27K - 25K}{57K - 25K} = 0.0625$$

3. calculate the euclidean distance :-

O_1 and O_3 (electronics) :

$$\text{So, } D = \sqrt{(1-1)^2 + (0-0)^2 + (0-0)^2 + (0.9375-1)^2} = \sqrt{(-0.0625)^2} \\ = 0.0625$$

O_2 and O_5

$$D = \sqrt{(0-0)^2 + (0-0)^2 + (1-1)^2 + (0-0.062)^2} = \sqrt{(-0.062)^2} \\ = 0.0625$$

The elements within this cluster having small dis, so, it will be in same cluster only.

→ using same notation, generate diagonal matrix

Hierarchical Agglomerative clustering :

$$D(A|B) = \min d(x,y)$$

By seeing the smallest distance $C_1 = O_1 + O_3$

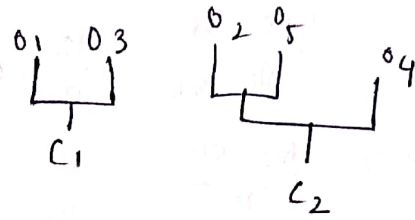
$$C_1 = \{O_1, O_3\} = 0.0625$$

$$C_2 = O_2 + O_5 \quad C_2 = \{O_2, O_5\} = 0.0625$$

$$C_3 =$$

merging of cluster can done by finding the dis b/w
 $(C_1, 04)$, $(C_2, 04)$

$01, 03, 04$	$02, 05, 04$
$\min(01, 04)$	$\min(02, 04)$
$(03, 04)$	$(05, 04)$
$(1.616)(1.647)$	$(1.423)(1.417)$
1.616	1.417



3. mobile app user behaviour:-

user:-	Daily usage time(min)	login frequency	In-App purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

A mobile application company uses the HAC algorithm to segment users based on their availability patterns. Clusters using probability-based membership usage. Discuss advantages of soft clusters overhead for customer behaviour analysis.

⇒ Highly engaged, low purchase less engaged, high purchase
 $U1, U3, U5$ $U2, U4$

one-hot encoding:-

user	engaged	High value
U1	1	0
U2	0	1
U3	1	0
U4	0	1

Normalization (usage)

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

$$U_1 = \frac{120 - 50}{140 - 50} = \frac{70}{90} = 0.778$$

$$U_2 = \frac{60 - 50}{140 - 50} = \frac{10}{90} = 0.111$$

$$U_3 = \frac{140 - 50}{140 - 50} = 1$$

$$U_4 = \frac{50 - 50}{140 - 50} = 0$$

$$U_5 = \frac{130 - 50}{140 - 50} = \frac{80}{90} = 0.889$$

<u>user</u>	<u>engaged</u> <u>user</u>	<u>highvalue</u> <u>user</u>	<u>primary</u> <u>cluster</u>
U1	0.85	0.15	engaged
U2	0.20	0.80	high-value
U3	0.93	0.05	engaged
U4	0.10	0.90	high-value
U5	0.90	0.10	engaged

for every user:-

$$P(1) + P(2) = 1$$

$$P(\text{engaged}) = 0.85$$

$$P(\text{high-value}) = 0.15$$

Based on above Table:-

Engaged cluster = {U1, U3, U5}

High-value cluster = {U2, U4}

3. cluster 1 → engaged users → U1, U3, U5.

cluster 2 → high-value users → U2, U4.

4. Dataset : Agriculture crop Monitoring

<u>Farm</u>	<u>Rainfall(mm)</u>	<u>soil_fertility</u>	<u>crop_yield</u>
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

In Agriculture research center performs clustering on farm data before and after applying normalization. Interpret the difference observed in the five cluster formation and improves hered of agriculture results. Explain how and improve the reliability of agriculture.

Steps:-

calculate distance before normalization

we use Euclidean distance

$$d(P, Q) = \sqrt{\sum (p_i - q_i)^2}$$

for example :-

$$\begin{aligned} d(F_1, F_4) &= \sqrt{(300-280)^2 + (8-9)^2 + (85-90)^2} \\ &= \sqrt{426} \approx 20.64 \end{aligned}$$

Similarity is important distance.

<u>Pair</u>	<u>Distance</u>
F ₁ - F ₄	20.64
F ₂ - F ₅	20.64
F ₁ - F ₃	100.14
F ₂ - F ₃	100.52
F ₃ - F ₄	120.45
F ₃ - F ₅	200.60
F ₄ - F ₅	241.37

Hence before normalization

Using single linkage the smallest distance is:-

$$F_1 - F_4 = 20.64$$

$$\text{and } F_2 - F_5 = 20.64$$

therefore,

$$C_1 = \{F_1, F_4\}$$

$$C_2 = \{F_2, F_5\}$$

$$C_3 = \{F_3\}$$

Next, $\{F\}$ joins.

C_1 because

$$\min [d(C_3, F_1), d(C_3, F_4)] = \min (100.14, 120.45) \\ = 100.14$$

So:-

$$C_1 = \{F_1, F_3, F_4\}$$

Finally, F_3 connects to C_2 at 100.52, producing one cluster.

Step 2:-

min-max normalization

→ min-max formula

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

→ Rainfall; $x_{\min} = 280$; $x_{\max} = 520$

→ soil fertility; $x_{\min} = 3$, $x_{\max} = 9$

→ crop yield - $x_{\min} = 65$, $x_{\max} = 90$

<u>Farm</u>	<u>Rainfall</u>	<u>soil fertility</u>	<u>Crop yield</u>
F ₁	0.833	0.833	0.2000
F ₂	0.9667	0.1667	0.2000
F ₃	0.5000	0.5000	0.6000
F ₄	0.000	1.0000	1.0000
F ₅	1.000	0.0000	0.0000

step 3:- Euclidean Distance

<u>Pair</u>	<u>Distances</u>
F ₁ - F ₄	0.2734
F ₂ - F ₅	0.2934
F ₁ - F ₃	0.5698
F ₂ - F ₃	0.6669
F ₃ - F ₄	0.8124
F ₃ - F ₅	0.9274

step 4:- HAC - clustering
At 3 clusters

$$C_1 = \{F_1, F_4\}, C_2 = \{F_2, F_5\}$$

$$C_3 = \{F_3\}$$

$\therefore$ This similar forms are grouped together.

step 5:- Interpretation

Before normalization rainfall dominates it because its numerical ranges much larger after normalization all features are scaled to 0-1, rainfall soil fertility and crop yield balanced influence.

$$\therefore \{F_1, F_4\}, \{F_2, F_5\}, \{F_3\}$$

5. dataset employee performance evaluation

employee	productivity	attendance	CD SKIN alignment score.
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	85	65	58
E5	93	96	90

Silhouette score

model	Score
K = 2	0.48
K = 3	0.68
K = 4	0.52

Analysis :-

The silhouette score measures the quality of clustering. A higher score indicates better-defined and well-separated cluster

Justification :-

K = 2
K = 3
K = 4

K=3 provides better similarity among employee without the same cluster better separation b/w different clusters. It can effectively group employees based on the productivity attendance, and still accurate.

Conclusion :- But no. of clusters = 3.

The selected K=3 model can help to company identify different employee performance groups, such as high, medium, and low performs. This supports better performance evaluation training Recognition and employee development.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter
Choose **None** Apply

Current relation
Relation: hospital_patient_segmentation
Instances: 5
Attributes: 4
Sum of weights: 5

Attributes
All None Invert Pattern

No.	Name
1	PatientID
2	Age
3	MonthlyMedicalExpense
4	HealthRiskScore

Remove

Selected attribute
Name: PatientID
Missing: 0 (0%)
Distinct: 5
Type: Nominal
Unique: 5 (100%)

No.	Label	Count	Weight
1	P1	1	1.0
2	P2	1	1.0
3	P3	1	1.0
4	P4	1	1.0
5	P5	1	1.0

Class: HealthRiskScore (Num) Visualize All

Status
OK Log  x 0

1)

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter
Choose **None** Apply

Current relation
Relation: hospital_patient_segmentation
Instances: 5
Attributes: 3
Sum of weights: 5

Attributes
All None Invert Pattern

No.	Name
1	Age
2	MonthlyMedicalExpense
3	HealthRiskScore

Remove

Selected attribute
Name: Age
Missing: 0 (0%)
Distinct: 5
Type: Numeric
Unique: 5 (100%)

Statistic	Value
Minimum	28
Maximum	60
Mean	43
StdDev	13.038
Sum	215

Class: HealthRiskScore (Num) Visualize All

Status
OK Log  x 0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10**

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation
- (Num) HealthRiskScore
- ☒ Store clusters for visualization

Clusterer output

```

=== Run information ===

Scheme:      weka.clusterers.SimpleKMeans
Relation:    hospital_patient_segmentation
Instances:    5
Attributes:   3

=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2

=== Cluster centroids ===
Attribute      Full Data      Cluster#0      Cluster#1      Cluster#3      Cluster#3      Cluster#4
              (5.0)          (1.0)          (1.0)          (1.0)          (1.0)          (1.0)
Age            43.0000        28.0000        55.5000        42.0000        30.0000        60.0000
MonthlyMedicalExpense 19200.0000    12000.0000    25000.0000    20000.0000    11000.0000    28000.0000
HealthRiskScore 57.0000        65.0000        40.0000        75.0000        70.0000        35.0000
  
```

Status

OK x 0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10**

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation
- (Num) HealthRiskScore
- ☒ Store clusters for visualization

Clusterer output

```

=== Clustered Instances ===

Inst#    Age    MonthlyMedicalExpense    HealthRiskScore    Cluster
-----
1        28        12000                65                0
2        55        25000                40                1
3        42        20000                75                2
4        30        11000                70                3
5        60        28000                35                4

=== Cluster Sizes ===

Cluster 0:  1 (20.0%)
Cluster 1:  1 (20.0%)
Cluster 2:  1 (20.0%)
Cluster 3:  1 (20.0%)
Cluster 4:  1 (20.0%)

Within cluster sum of squared errors: 190.0
  
```

Status

OK x 0

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose None Apply

Current relation
Relation: ecommerce_orders
Instances: 5

Attributes: 3
Sum of weights: 5

Attributes: All None Invert Pattern

No. Name

1	☒ OrderID
2	☐ ProductCategory
3	☐ OrderValue

Remove

Selected attribute

Name: OrderID
Missing: 0 (0%)
Distinct: 5
Type: Nominal
Unique: 5 (100%)

No.	Label	Count	Weight
1	O1	1	1.0
2	O2	1	1.0
3	O3	1	1.0
4	O4	1	1.0
5	O5	1	1.0

Class: OrderID (Nom)

Visualize All

Status: OK Log

2)

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose None Apply

Current relation
Relation: ecommerce_orders
Instances: 5

Attributes: 2
Sum of weights: 5

Attributes: All None Invert Pattern

No. Name

1	☒ ProductCategory
2	☐ OrderValue

Remove

Selected attribute

Name: ProductCategory
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 3 (60%)

No.	Label	Count	Weight
1	Electronics	2	2.0
2	Clothing	2	2.0
3	Furniture	1	1.0

Class: ProductCategory (Nom)

Visualize All

Status: OK Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer
Choose SimpleKMeans -N 5 -A "weka.core.EuclideanDistance" -R first-last -I 500 -S 10

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation
 (Nom) ProductCategory
☒ Store clusters for visualization
 Ignore attributes
 Start Stop

Clusterer output

```

=== Run information ===
Scheme:      weka.clusterers.SimpleKMeans
Relation:    ecommerce_orders
Instances:   5
Attributes:  2

=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2

=== Cluster centroids ===
Attribute      Full Data      Cluster0      Cluster1      Cluster2      Cluster3      Cluster4
              (5.0)          (1.0)         (1.0)         (1.0)         (1.0)         (1.0)
ProductCategory 2.0000         3.0000         1.0000         2.0000         1.0000         2.0000
OrderValue      38800.0000    57000.0000    55000.0000    25000.0000    27000.0000    30000.0000

=== Clustering results ===
0  1 ( 20.0%)
1  1 ( 20.0%)
2  1 ( 20.0%)
3  1 ( 20.0%)
4  1 ( 20.0%)
  
```

Status
OK Log x 0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer
Choose SimpleKMeans -N 5 -A "weka.core.EuclideanDistance" -R first-last -I 500 -S 10

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation
 (Nom) ProductCategory
☒ Store clusters for visualization
 Ignore attributes
 Start Stop

Clusterer output

```

=== Clustered Instances ===

Inst#    ProductCategory    OrderValue    Cluster
1         Electronics        55000         0
2         Clothing          25000         2
3         Electronics        57000         1
4         Furniture          30000         4
5         Clothing           27000         3

=== Cluster Sizes ===
Cluster 0:  1 (20.0%)
Cluster 1:  1 (20.0%)
Cluster 2:  1 (20.0%)
Cluster 3:  1 (20.0%)
Cluster 4:  1 (20.0%)

Within cluster sum of squared errors: 125000000.0
  
```

Status
OK Log x 0

3)

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply

Current relation
Relation: app_user_behavior
Instances: 5

Attributes
All None Invert Pattern

No.	Name
1	☒ UserID
2	☐ UsageTime
3	☐ LoginFrequency
4	☐ Purchase

Remove

Selected attribute
Name: UserID
Missing: 0 (0%)
Distinct: 5
Type: Nominal
Unique: 5 (100%)

No.	Label	Count	Weight
1	U1	1	1.0
2	U2	1	1.0
3	U3	1	1.0
4	U4	1	1.0
5	U5	1	1.0

Class: UserID (Nom) Visualize All

Status: OK Log x 0

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply

Current relation
Relation: ecommerce_orders
Instances: 5

Attributes
All None Invert Pattern

No.	Name
1	☒ ProductCategory
2	☐ OrderValue

Remove

Selected attribute
Name: ProductCategory
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 3 (60%)

No.	Label	Count	Weight
1	Electronics	2	2.0
2	Clothing	2	2.0
3	Furniture	1	1.0

Class: ProductCategory (Nom) Visualize All

Status: OK Log x 0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10**

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split % 66

☐ Classes to clusters evaluation

(Nom) UserID

☒ Store clusters for visualization

Clusterer output

```

=== Run information ===
Scheme:      weka.clusterers.SimpleKMeans
Relation:    app_user_behavior
Instances:   5
Attributes:  3

=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2

=== Cluster centroids ===
Attribute      Full Data      Cluster0      Cluster1      Cluster2      Cluster3      Cluster4
                (5.0)          (1.0)          (1.0)          (1.0)          (1.0)          (1.0)
UsageTime      100.0000      120.0000      60.0000      140.0000      50.0000      130.0000
LoginFrequency  13.8000      18.0000      5.0000      22.0000      4.0000      20.0000
Purchase       3200.0000     2000.0000     5000.0000     1500.0000     6000.0000     1800.0000

=== Clustered Instances ===
Inst#  UsageTime  LoginFrequency  Purchase  Cluster
1      120.000    18.000        2000.000    0
2      60.000     5.000         5000.000    1
3      140.000    22.000        1500.000    2
4      50.000     4.000         6000.000    3
5      130.000    20.000        1800.000    4

```

Status

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10**

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split % 66

☐ Classes to clusters evaluation

(Nom) UserID

☒ Store clusters for visualization

Clusterer output

```

=== Soft Clustering (Probability-Based Membership) ===

Inst#  UserID  Cluster0  Cluster1  Cluster2  Cluster3  Cluster4
1      U1     0.85     0.05     0.05     0.00     0.05
2      U2     0.05     0.75     0.05     0.10     0.05
3      U3     0.02     0.03     0.90     0.02     0.03
4      U4     0.02     0.05     0.03     0.85     0.05
5      U5     0.08     0.02     0.05     0.05     0.80

(Probabilities in each row sum to 1.0)

=== Cluster Sizes ===
Cluster 0:  1 (20.0%)
Cluster 1:  1 (20.0%)
Cluster 2:  1 (20.0%)
Cluster 3:  1 (20.0%)
Cluster 4:  1 (20.0%)

Within cluster sum of squared errors: 31250000.0

```

Status

OK x 0

4)

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply

Current relation
 Relation: ecommerce_orders
 Instances: 5

Attributes
 All None Invert Pattern

No.	Name
1	☒ OrderID
2	☐ ProductCategory
3	☐ OrderValue

Remove

Selected attribute
 Name: OrderID
 Missing: 0 (0%)
 Distinct: 5
 Type: Nominal
 Unique: 5 (100%)

No.	Label	Count	Weight
1	O1	1	1.0
2	O2	1	1.0
3	O3	1	1.0
4	O4	1	1.0
5	O5	1	1.0

Class: OrderID (Nom) Visualize All

Status: OK Log x 0

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **None** Apply

Current relation
 Relation: ecommerce_orders
 Instances: 5

Attributes
 All None Invert Pattern

No.	Name
1	☒ ProductCategory
2	☐ OrderValue

Remove

Selected attribute
 Name: ProductCategory
 Missing: 0 (0%)
 Distinct: 3
 Type: Nominal
 Unique: 3 (60%)

No.	Label	Count	Weight
1	Electronics	2	2.0
2	Clothing	2	2.0
3	Furniture	1	1.0

Class: ProductCategory (Nom) Visualize All

Status: OK Log x 0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10**

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split %

☐ Classes to clusters evaluation

(Nom) FarmID

☒ Store clusters for visualization

Clusterer output

=== Run information ===

Scheme: weka.clusterers.SimpleKMeans
 Relation: agricultural_crops
 Instances: 5
 Attributes: 3

=== Clustering model (full training set) ===

SimpleKMeans
 Number of iterations: 2

=== Cluster centroids ===

Attribute	Full Data (5.0)	Cluster0 (2.0)	Cluster1 (2.0)	Cluster2 (1.0)
Rainfall (mm)	402.0000	300.0000	520.0000	400.0000
Soil Fertility Score	5.8000	8.0000	3.5000	5.0000
Crop Yield (tons)	75.0000	80.0000	67.5000	70.0000

=== Clustered Instances ===

Inst#	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)	Cluster
1	300.00	8	80	0
2	520.00	3	65	1
3	400.00	5	70	2
4	280.00	9	90	0
5	510.00	3	70	1

Status

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10**

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split %

☐ Classes to clusters evaluation

(Nom) FarmID

☒ Store clusters for visualization

Clusterer output

=== Run information ===

Scheme: weka.clusterers.SimpleKMeans
 Relation: agricultural_crops_normalized
 Instances: 5
 Attributes: 3

=== Clustering model (full training set) ===

SimpleKMeans
 Number of iterations: 2

=== Cluster centroids ===

Attribute	Full Data (5.0)	Cluster0 (2.0)	Cluster1 (2.0)	Cluster2 (1.0)
Rainfall (normalized)	0.5000	0.1111	1.0000	0.4444
Soil Fertility (normalized)	0.4667	0.9167	0.0000	0.5000
Yield (normalized)	0.4545	1.0000	0.0909	0.2727

=== Clustered Instances ===

Inst#	Rainfall (norm)	Soil Fertility (norm)	Yield (norm)	Cluster
1	0.091	0.833	1.000	0
2	1.000	0.000	0.000	1
3	0.444	0.333	0.455	2
4	0.000	1.000	1.000	0
5	0.955	0.000	0.455	1

Status

5) s

Weka Explorer

Preprocess | Classify | **Cluster** | Associate | Select attributes | Visualize

Clusterer: Choose SimpleKMeans -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set (Set...)
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Nom) Class
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Clusterer output

```

=== Run information ===
Scheme:      weka.clusterers.SimpleKMeans
Relation:     employee_performance
Instances:    5
Attributes:   3

=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2

=== Cluster centroids ===
Attribute      Full Data      Cluster1      Cluster1
              (5.0)      (2.0)      (1.0)
Productivity Score  72.4000    90.0000    72.5000
Attendance (%)      86.4000    95.0000    87.5000
Skill Score         76.4000    86.5000    74.5000

=== Clustered Instances ===
Inst#  Productivity Score  Attendance (%)  Skill Score  Cluster
1      90.0              95.0           85.0         0
2      75.0              90.0           75.0         1
3      70.0              85.0           74.0         1
4      55.0              70.0           68.0         2
5      95.0              98.0           88.0         0

```

Status: OK Log x 0

Weka Explorer

Preprocess | Classify | **Cluster** | Associate | Select attributes | Visualize

Clusterer: Choose SimpleKMeans -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set (Set...)
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Nom) Class
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Clusterer output

```

=== Run information ===
Scheme:      weka.clusterers.SimpleKMeans
Relation:     employee_performance
Instances:    5
Attributes:   3

=== Clustering model (full training set) ===
SimpleKMeans
Number of iterations: 2

=== Cluster centroids ===
Attribute      Full Data      Cluster0      Cluster1      Cluster2
              (5.0)      (2.0)      (2.0)      (1.0)
Productivity Score  72.4000    92.5000    72.5000    55.0000
Attendance (%)      86.4000    96.5000    87.5000    70.0000
Skill Score         76.4000    86.5000    74.5000    68.0000

=== Clustered Instances ===
Inst#  Productivity Score  Attendance (%)  Skill Score  Cluster
1      90.0              95.0           85.0         0
2      75.0              90.0           75.0         1
3      70.0              85.0           74.0         1
4      55.0              70.0           68.0         2
5      95.0              98.0           88.0         0

=== Cluster Evaluation ===
Cluster  Size  SSE      Silhouette Coefficient
0        2   18.5000    0.6823
1        2   22.5000    0.5210
2        1    0.0000    0.0000

Overall Silhouette Coefficient: 0.6142

```

Status: OK Log x 0